

Jaden Toh

+65 86668412 | jadentyh@gmail.com | linkedin.com/in/jadentyh | github.com/JadenTyh

EDUCATION

Singapore Management University

Aug 2023 – Expected May 2027

Bachelor of Science (Computer Science) – AI & Software Systems track

- **Relevant Coursework:** Data Structures and Algorithms, Database Systems, DevOps, Machine Learning

EXPERIENCE

A*STAR | Research Engineer | *Python, MLflow, FastAPI, Scikit-learn, Pydantic*

Jan 2026 – Present

- **Time-Series Forecasting:** Prevented lookahead bias in forecasting pipelines by enforcing strict temporal train-test splits, causal LOESS smoothing, and split-integrity validation via regression tests.
- **MLOps:** Enabled end-to-end experiment traceability and streamlined model deployment by integrating MLflow to log, version, and retrieve forecasting models and metrics.
- **FastAPI:** Built a FastAPI experiment engine with asynchronous background tasks for concurrent multi-experiment submissions, adding lifespan-managed initialization and fail-fast error handling for reliable multi-model orchestration.

Hontrel | AI Engineer | *Python, GraphQL, FastMCP*

May 2025 – Dec 2025

- Built Agentic GitHub Project Manager assistant that turns natural-language requests into GitHub Project Actions
- Improved reliability and maintainability by moving from a single “tool handler” design to a multi-agent pipeline, resulting in ~50% faster feature iteration and ~30% fewer runtime failures.
- Exposed a single high-level MCP tool via a FastMCP server to integrate seamlessly with Claude Desktop.
- Improved intent detection accuracy by 30% by using Claude Desktop over traditional manual ML systems.

PROJECTS

Noise Detector for User-Rating-Movies | *Python, Pandas, Scikit-learn*

Oct 2025 – Nov 2025

- Architected a multi-stage stacked ensemble (LightGBM, Random Forest, Ridge Meta-Learner) that filtered high-variance noise in user rating data, achieving a top-tier 0.062 MAE on unseen test data.
- Designed a 5-fold Out-of-Fold (OOF) cross-validation pipeline to prevent data leakage and ensure robust model validation under noisy data conditions.

RoadStar Project | *Kubernetes, AWS, ArgoCD, Next.js, TypeScript*

Oct 2025 – Present

- Delivered a modern, modular travel planning application by implementing a Microservices Architecture with Kubernetes and cloud infrastructure as code.
- Established end-to-end DevOps foundations and CI/CD pipelines as Infrastructure Lead in a team of 7, achieving consistent security and deployment standards across four services.
- Achieved Continuous Deployment on Amazon EKS by leveraging ArgoCD for GitOps-based delivery.
- Will be the Frontend lead in part 2, building a Next.js/TypeScript client that integrates with the four microservices and JWT-based auth system delivered in Part 1.

TECHNICAL SKILLS

Languages: Python, Java, C, JavaScript, HTML, CSS, SQL

Tools & Platforms: Docker, Git, AWS, Kubernetes, MongoDB, GitHub GraphQL API, FastAPI, Pydantic, NeuralForecast

Frameworks: Spring Boot, Flask, Matplotlib, Seaborn, Bootstrap, NumPy, MLflow, FastMCP

CERTIFICATIONS & COMPETITIONS

AWS Certified Solutions Architect – Associate: Solution Architecture

Fortinet NSE Certifications: Fundamentals of Cyber Security & Threat Landscape

Elipsis Code 2024 (March 2024): Ranked top 50 in a competitive coding competition hosted by SMU Ellipsis

ACTIVITIES & LEADERSHIP

SMU Tennis: Elected as Logistics Director (2024) and Men’s Captain (2025); represented SMU in Singapore University Games 2023, 2024, 2025 & 2026.